Multiple Linear Regression

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From SLR to MLR

Recap: simple linear regression

The model with one regressor:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n.$$

OLS estimators of the coefficients, with $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum_i (x_i - \bar{x})^2$, and the unbiased estimator of the error variance:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \tag{1}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

$$\hat{\sigma}^2 = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-2}. \tag{2}$$

Question: is one regressor enough?

Delivery time of a soft-drink route driver depends on the number of cases loaded (x_1) and the walking distance (x_2).
Why not simply fit two separate SLRs, y on x_1 and y on x_2?

- The two regressors are typically correlated (more cases usually means longer distance).
- Each SLR slope then absorbs part of the effect of the omitted regressor, so it is not the effect of that regressor alone.
- Separate fits cannot answer “what is the effect of one more case for a fixed distance?”.
- The residual variance is also larger because the omitted regressor’s effect is left in the error.

Example: delivery time data

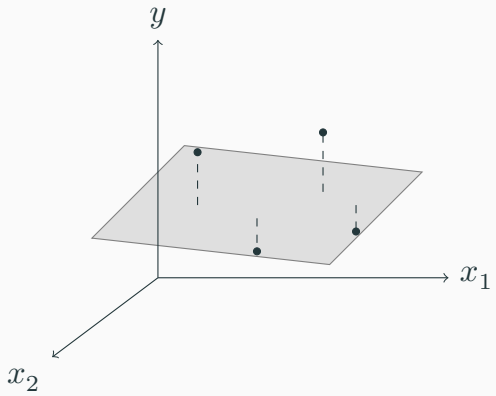
Montgomery, Peck & Vining, Example 3.1 ($n = 25$).

- y: delivery time (minutes)
- x_1: number of cases stocked
- x_2: distance walked by the driver (feet)

Obs.	Time y	Cases x_1	Distance x_2
1	16.68	7	560
2	11.50	3	220
3	12.03	3	340
⋮	⋮	⋮	⋮
25	10.75	4	150

From a line to a plane

- SLR: a line in the (x, y) plane.
- Two regressors: a plane in (x_1, x_2, y) space.
- k regressors: a hyperplane in $(k + 1)$ dimensions.
- Residuals: vertical distances to the fitted surface.



The MLR model

The MLR equation

With k regressors and n observations:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, \ldots, n. \tag{3}$$

- x_{ij}: value of regressor j for observation i.
- $p = k + 1$ unknown regression coefficients $\beta_0, \ldots, \beta_k$ (including the intercept).
- ε_i: random error of observation i.

Meaning of the coefficients

$$\frac{\partial \, \mathrm{E}[y \mid x_1, \ldots, x_k]}{\partial x_j} = \beta_j, \quad j = 1, \ldots, k.$$

- β_j is the change in the mean response per unit increase in x_j *when all other regressors are held fixed*.
- For this reason $\beta_1, \ldots, \beta_k$ are called partial regression coefficients.
- β_0 is the mean response when all regressors are zero (meaningful only if that point is within the range of the data).

Question: interpreting the coefficients

In the delivery time model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$, interpret β_1, β_2 and β_0 in words, with units.

From n equations to one matrix equation

Writing (3) out for every observation:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_{11} + \cdots + \beta_k x_{1k} + \varepsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_{21} + \cdots + \beta_k x_{2k} + \varepsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_{n1} + \cdots + \beta_k x_{nk} + \varepsilon_n \end{aligned}$$

Stack the responses, the coefficients and the errors into vectors, and the regressor values into a matrix.

The MLR model in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{4}$$

$$\underset{n\times 1}{\underset{\sim}{\mathbf{y}}}\,, \qquad \underset{n\times p}{\underset{\sim}{\mathbf{X}}}\,, \qquad \underset{p\times 1}{\underset{\sim}{\boldsymbol{\beta}}}\,, \qquad \underset{n\times 1}{\underset{\sim}{\boldsymbol{\varepsilon}}}\,.$$

The vectors $\mathbf{y}$, $\boldsymbol{\beta}$, $\boldsymbol{\varepsilon}$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \qquad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

The design matrix $\mathbf{X}$

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}$$

- Row i is observation i; column j is regressor j.
- The first column of ones corresponds to the intercept β_0.

Model assumptions

$$\mathrm{E}[\boldsymbol{\varepsilon}] = \mathbf{0}, \tag{5}$$
$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n. \tag{6}$$

- $\mathbf{X}$ is fixed (non-random) and has full column rank, i.e. $\mathrm{rank}(\mathbf{X}) = p \leq n$.
- No distributional assumption on $\boldsymbol{\varepsilon}$ is needed yet for estimation (however noramility assumption is required later for inference)

$\mathrm{E}[\boldsymbol{\varepsilon}]$ is a n dimensional vector of zeroes.

What does $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ say?

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \mathrm{E}[(\boldsymbol{\varepsilon} - \mathbf{0})(\boldsymbol{\varepsilon} - \mathbf{0})'] = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix}$$

- $\mathrm{Var}(\boldsymbol{\varepsilon})$ is $n \times n$ dimensional variance-covariance matrix.
- Diagonal entries: the i-th diagonal entry contains the variance of the i-th error term. In this model, every error has the same variance σ^2 (constant variance).
- Off-diagonal entries: cell (i, l) contains the covariance between observations i and l. In this model, errors of different observations are uncorrelated, hence covariances are zero.

“Linear” means linear in the parameters

Each of these is an MLR model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon,$$
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon.$$

This one is not:

$$y = \beta_0 e^{\beta_1 x} + \varepsilon.$$

Polynomial and interaction terms are handled later in the course.

OLS estimation in matrix form

The least-squares criterion

Choose $\boldsymbol{\beta}$ to minimise the sum of squared errors

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{n} \varepsilon_i^2 = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \tag{7}$$

Expanding $S(\boldsymbol{\beta})$

$$\begin{aligned} S(\boldsymbol{\beta}) &= \mathbf{y}'\mathbf{y} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \\ &= \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}. \end{aligned} \tag{8}$$

The second line combines the two middle terms.

Why is $\mathbf{y}'\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{X}'\mathbf{y}$?

Two rules for differentiating with respect to a vector

For a constant vector $\mathbf{a}$ and a constant symmetric matrix $\mathbf{A}$:

$$\frac{\partial\, \mathbf{a}'\boldsymbol{\beta}}{\partial \boldsymbol{\beta}} = \mathbf{a}, \qquad \frac{\partial\, \boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}}{\partial \boldsymbol{\beta}} = 2\mathbf{A}\boldsymbol{\beta}.$$

Apply with $\mathbf{a} = \mathbf{X}'\mathbf{y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$ (symmetric) to (8).

The normal equations

$$\begin{aligned}
\frac{\partial S}{\partial \boldsymbol{\beta}} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta}, \\
\left.\frac{\partial S}{\partial \boldsymbol{\beta}}\right|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}, \\
\Rightarrow \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} &= \mathbf{X}'\mathbf{y}. \qquad (9)
\end{aligned}$$

This is a system of p linear equations in the p unknowns $\hat{\beta}_0, \ldots, \hat{\beta}_k$.

The OLS estimator

If $\mathbf{X}'\mathbf{X}$ is invertible, the normal equations (9) have the unique solution

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}. \tag{10}$$

When is $\mathbf{X}'\mathbf{X}$ invertible?

- $\mathbf{X}'\mathbf{X}$ ($p \times p$) is invertible if and only if $\mathbf{X}$ has full column rank.
- That requires $n \geq p$ and no column of $\mathbf{X}$ being a linear combination of the other columns.
- If a regressor is (nearly) a linear combination of the others, $\mathbf{X}'\mathbf{X}$ is singular (nearly singular): this is the collinearity problem, discussed in a later lecture.

Is $\hat{\beta}$ a minimum?

$$\frac{\partial^2 S}{\partial \boldsymbol{\beta}\, \partial \boldsymbol{\beta}'} = 2\mathbf{X}'\mathbf{X}.$$

- For any $\mathbf{a} \neq \mathbf{0}$: $\mathbf{a}'\mathbf{X}'\mathbf{X}\mathbf{a} = (\mathbf{X}\mathbf{a})'(\mathbf{X}\mathbf{a}) = \|\mathbf{X}\mathbf{a}\|^2 > 0$, because full column rank gives $\mathbf{X}\mathbf{a} \neq \mathbf{0}$.
- So $\mathbf{X}'\mathbf{X}$ is positive definite and $\hat{\boldsymbol{\beta}}$ is the unique minimiser of (7).

Fitted values, residuals and the hat matrix

Fitted values and residuals

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$
$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y},$$
$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \tag{11}$$
$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}. \tag{12}$$

$\mathbf{H}$ is the $n \times n$ hat matrix: it puts the hat on $\mathbf{y}$.

Properties of the hat matrix

$$\mathbf{H}' = \mathbf{H}, \tag{13}$$
$$\mathbf{H}^2 = \mathbf{H}, \tag{14}$$
$$\mathbf{HX} = \mathbf{X}. \tag{15}$$

That is, $\mathbf{H}$ is symmetric and idempotent.

Prove (14) and (15) from the definition (11).

Properties of $\mathbf{I} - \mathbf{H}$

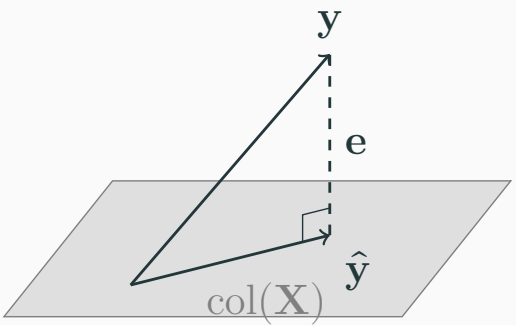
From (13), (14) and (15):

$$(\mathbf{I} - \mathbf{H})' = \mathbf{I} - \mathbf{H}, \tag{16}$$
$$(\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - \mathbf{H}, \tag{17}$$
$$(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}. \tag{18}$$

Geometry of least squares

$\mathbf{H}$ is symmetric and idempotent and $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, therefore, $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathbf{X}$.

- The columns of $\mathbf{X}$ span a p-dimensional subspace $\text{col}(\mathbf{X})$ of $\mathbb{R}^n$.
- $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is the orthogonal projection of $\mathbf{y}$ onto $\text{col}(\mathbf{X})$.
- $\mathbf{e}$ is perpendicular to $\text{col}(\mathbf{X})$.
- $\hat{\mathbf{y}}$ is the point of $\text{col}(\mathbf{X})$ closest to $\mathbf{y}$, so $\hat{\beta}$ minimises $\|\mathbf{y} - \mathbf{X}\beta\|^2$.



Orthogonality of residuals and regressors

$$\mathbf{X}'\mathbf{e} = \mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}, \tag{19}$$

by the normal equations (9). Consequences:

- First row of $\mathbf{X}'$ is a row of ones, so $\sum_i e_i = 0$.
- Row j gives $\sum_i x_{ij} e_i = 0$ for each regressor.
- $\hat{\mathbf{y}}'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = 0$.

Delivery time example: OLS fit

Computing $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{y}$

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 25 & 219 & 10\,232 \\ 219 & 3\,055 & 133\,899 \\ 10\,232 & 133\,899 & 6\,725\,688 \end{pmatrix},$$

$$\mathbf{X}'\mathbf{y} = \begin{pmatrix} 559.60 \\ 7\,375.44 \\ 337\,071.69 \end{pmatrix}.$$

The inverse

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 1.13 \times 10^{-1} & -4.45 \times 10^{-3} & -8.37 \times 10^{-5} \\ -4.45 \times 10^{-3} & 2.74 \times 10^{-3} & -4.79 \times 10^{-5} \\ -8.37 \times 10^{-5} & -4.79 \times 10^{-5} & 1.23 \times 10^{-6} \end{pmatrix}$$

The fitted equation

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \begin{pmatrix} 2.3412 \\ 1.6159 \\ 0.01438 \end{pmatrix}$$

$$\hat{y} = 2.3412 + 1.6159\, x_1 + 0.01438\, x_2.$$

Interpreting the fit

- Each additional case adds about 1.62 minutes to the delivery time, for a fixed distance.
- Each additional foot adds about 0.0144 minutes (about 1.4 minutes per 100 ft), for a fixed number of cases.
- The SLR of y on x_1 alone gives a slope of 2.18, not 1.62: cases and distance are correlated ($r = 0.82$), and the SLR slope absorbs part of the distance effect.

Verifying in R

```
y <- delivery$time
X <- cbind(1, delivery$cases, delivery$distance)

solve(t(X) %*% X, t(X) %*% y)

coef(lm(time ~ cases + distance,
        data = delivery))
```

- `solve(A, b)` solves $\mathbf{A}\mathbf{x} = \mathbf{b}$ directly; it is numerically preferable to forming $\mathbf{A}^{-1}$ explicitly.

Properties of the OLS estimators

Linearity

$$\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y}, \qquad \mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \tag{20}$$

$\mathbf{A}$ is a $p \times n$ matrix of constants (given $\mathbf{X}$), so every $\hat{\beta}_j$ is a linear combination of $y_1, \ldots, y_n$.

Decomposing $\hat{\beta}$

Substitute the model (4) into (10):

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ \hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}. \end{aligned} \tag{21}$$

The estimator is the true $\boldsymbol{\beta}$ plus a linear function of the errors.

Unbiasedness

Taking expectations in (21) with $\mathbf{X}$ fixed, and using (5):

$$\begin{aligned}\mathrm{E}[\hat{\boldsymbol{\beta}}] &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathrm{E}[\boldsymbol{\varepsilon}] \\ &= \boldsymbol{\beta}.\end{aligned}$$

The covariance matrix of a random vector

$\hat{\boldsymbol{\beta}}$ is p dimensional random vector. Therefore, $\mathrm{Var}(\hat{\boldsymbol{\beta}})$ is a $p \times p$ symmetric matrix:

- diagonal entries: $\mathrm{Var}(\hat{\beta}_j)$;
- off-diagonal entries: $\mathrm{Cov}(\hat{\beta}_i, \hat{\beta}_j)$.

For a constant matrix $\mathbf{B}$ and a random vector $\mathbf{z}$, directly from the definition,

$$\mathrm{Var}(\mathbf{B}\mathbf{z}) = \mathbf{B}\ \mathrm{Var}(\mathbf{z})\, \mathbf{B}'. \tag{22}$$

Covariance matrix of $\widehat{\beta}$

$$
\begin{aligned}
\operatorname{Var}(\hat{\boldsymbol{\beta}}) &= \operatorname{Var}\left(\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\right) \\
&= \operatorname{Var}\left((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\right) \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\, \operatorname{Var}(\boldsymbol{\varepsilon})\, \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}. \qquad (23)
\end{aligned}
$$

The steps use (21) (the constant $\boldsymbol{\beta}$ adds no variance), (22) and (6).

Reading the covariance matrix

Let $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}$, with entries C_{ij}, $i, j = 0, \dots, k$.

$$\begin{aligned} \mathrm{Var}(\hat{\beta}_j) &= \sigma^2 C_{jj}, \\ \mathrm{Cov}(\hat{\beta}_i, \hat{\beta}_j) &= \sigma^2 C_{ij}, \quad i \neq j. \end{aligned} \tag{24}$$

Question: what makes $\mathrm{Var}(\hat{\beta}_j)$ large?

Using (24), which features of the data and the design make the variance of $\hat{\beta}_j$ large?

- Large error variance σ^2.
- Small sample size n.
- Little spread in regressor $\mathbf{x}_j$ (as in SLR, $\mathrm{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$).
- Strong linear relationship between $\mathbf{x}_j$ and the other regressors: C_{jj} becomes large. This is the collinearity problem.

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Properties of the OLS estimators

Distribution of the estimates under normal errors

Distribution of $\hat{\beta}$

Add the assumption $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$. By (21), $\hat{\boldsymbol{\beta}}$ is a linear function of a normal vector, hence normal:

$$\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta},\ \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}). \tag{25}$$

The mean and covariance matrix are the ones derived above. This result drives the inference procedures in the coming weeks.

The Gauss–Markov theorem

Gauss–Markov theorem

Under (5) and (6) (no normality needed), the OLS estimator $\hat{\boldsymbol{\beta}}$ is the best linear unbiased estimator (BLUE) of $\boldsymbol{\beta}$.

- Linear: a linear function of $\mathbf{y}$.
- Unbiased: $\mathrm{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$.
- Best: among all linear unbiased estimators, each $\hat{\beta}_j$, and more generally each $\mathbf{c}'\hat{\boldsymbol{\beta}}$, has the smallest variance.

Maximum Likelihood Estimation of MLR Parameters

Maximum likelihood under normal errors

The likelihood function is given by:

$$L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{2\sigma^2}\right)$$
$$= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right).$$

The log-likelihood of $(\boldsymbol{\beta}, \sigma^2)$ is

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \tag{26}$$

For any fixed σ^2, maximising over $\boldsymbol{\beta}$ is the same as minimising $S(\boldsymbol{\beta})$ in (7), so the ML estimator of $\boldsymbol{\beta}$ equals the OLS estimator.

Maximum likelihood estimator of σ^2

Setting $\partial \ell / \partial \sigma^2 = 0$ at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ gives

$$\hat{\sigma}^2_{\mathrm{ML}} = \frac{(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})}{n} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n}. \tag{27}$$

We show next that this estimator is biased.

Estimating the variance parameter σ^2

Residuals as a function of the errors

The residual vector can also be written as:

$$\begin{aligned}\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{H}\mathbf{y} &= (\mathbf{I} - \mathbf{H})\mathbf{y} \\ &= (\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}, . \end{aligned} \tag{28}$$

Equation (28) holds because $(\mathbf{I} - \mathbf{H})\mathbf{X} = 0$ as per (18). Then the residual sum of squares:

$$\begin{aligned}\mathrm{SS}_{\mathrm{Res}} = \sum_{i=1}^{n} e_i^2 &= \mathbf{e}'\mathbf{e} \\ &= \boldsymbol{\varepsilon}'(\mathbf{I} - \mathbf{H})'(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}. \end{aligned} \tag{29}$$

Equation (29) holds because $(\mathbf{I} - \mathbf{H})$ is symmetric and idempotent as per equations (16) and (17).

Expectation of a quadratic form

For a constant symmetric matrix $\mathbf{B}$:

$$\begin{aligned} \mathrm{E}[\boldsymbol{\varepsilon}'\mathbf{B}\boldsymbol{\varepsilon}] &= \mathrm{E}\left[\,\mathrm{tr}(\boldsymbol{\varepsilon}'\mathbf{B}\boldsymbol{\varepsilon})\right] = \mathrm{E}\left[\,\mathrm{tr}(\mathbf{B}\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}')\right] \\ &= \mathrm{tr}\left(\mathbf{B}\ \mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\right) = \sigma^2\ \mathrm{tr}(\mathbf{B}). \end{aligned} \tag{30}$$

- A scalar equals its trace.
- $\mathrm{tr}(\mathbf{MN}) = \mathrm{tr}(\mathbf{NM})$, and expectation ($\mathrm{E}$) commutes with tr.
- $\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \sigma^2\mathbf{I}_n$ becuase $\mathrm{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ as mentioned in (5) and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ as in (6), therefore, $\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}] = \mathrm{E}[(\boldsymbol{\varepsilon} - \mathbf{0})(\boldsymbol{\varepsilon} - \mathbf{0})'] = \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$.

The trace of $\mathbf{I} - \mathbf{H}$

$$\begin{aligned} \operatorname{tr}(\mathbf{H}) &= \operatorname{tr}\left(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right) = \operatorname{tr}\left((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\right) \\ &= \operatorname{tr}(\mathbf{I}_p) = p, \\ \operatorname{tr}(\mathbf{I}_n - \mathbf{H}) &= n - p. \end{aligned} \tag{31}$$

$\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}]$ and an unbiased estimator of σ^2

Applying (30) with $\mathbf{B} = \mathbf{I} - \mathbf{H}$ to (29), and using (31):

$$\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}] = \sigma^2(n-p), \tag{32}$$

$$\hat{\sigma}^2 = \mathrm{MS}_{\mathrm{Res}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-p}. \tag{33}$$

$\mathrm{E}[\mathrm{MS}_{\mathrm{Res}}] = \sigma^2$. For $p = 2$ this is the SLR estimator (2).

Question: why divide by $n - p$?

The n residuals are $e_i = y_i - \hat{y}_i$. Looking at (32), why is the divisor of SSE $n - p$ and not n? What is the bias of $\hat{\sigma}^2_{\mathrm{ML}}$ in (27)?

- The residuals satisfy the p linear constraints $\mathbf{X}'\mathbf{e} = \mathbf{0}$ (as seen in euqation 19), so only $n - p$ of them are free, therefore, $n - p$ degrees of freedom
- $\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}/n] = \sigma^2(n - p)/n$ which is smaller than σ^2
- So the ML estimator of the variance term, $\hat{\sigma}^2_{\mathrm{ML}} = \mathrm{SS}_{\mathrm{Res}}/n$ underestimates σ^2

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Standard errors of the coefficients

Replace σ^2 by $\hat{\sigma}^2$ in (23) and (24):

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1},$$
$$\mathrm{se}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}. \tag{34}$$

Delivery time data: $\hat{\sigma}^2$

- $\mathrm{SS}_{\mathrm{Res}} = 233.73$, with $n - p = 25 - 3 = 22$ degrees of freedom.
- $\hat{\sigma}^2 = \mathrm{MS}_{\mathrm{Res}} = 233.73/22 = 10.62$.
- $\hat{\sigma} = 3.26$ minutes: the typical deviation of a delivery time from the fitted plane.

Delivery time data: variance-covariance matrix of the estimates

From (23) with σ^2 replaced by $\hat{\sigma}^2 = 10.62$ and $(\mathbf{X}'\mathbf{X})^{-1}$ from before:

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1} = 10.62 \begin{pmatrix} 1.13 \times 10^{-1} & -4.45 \times 10^{-3} & -8.37 \times 10^{-5} \\ -4.45 \times 10^{-3} & 2.74 \times 10^{-3} & -4.79 \times 10^{-5} \\ -8.37 \times 10^{-5} & -4.79 \times 10^{-5} & 1.23 \times 10^{-6} \end{pmatrix}$$

$$\approx \begin{pmatrix} 1.20 & -4.73 \times 10^{-2} & -8.89 \times 10^{-4} \\ -4.73 \times 10^{-2} & 2.91 \times 10^{-2} & -5.09 \times 10^{-4} \\ -8.89 \times 10^{-4} & -5.09 \times 10^{-4} & 1.31 \times 10^{-5} \end{pmatrix}.$$

- The diagonal entries are $\widehat{\mathrm{Var}}(\hat{\beta}_j)$; their square roots are the standard errors 1.097, 0.171 and 0.00362 in (34).
- The off-diagonal entries are estimated covariances; e.g. $\widehat{\mathrm{Cov}}(\hat{\beta}_1, \hat{\beta}_2) = -5.09 \times 10^{-4} < 0$: the estimates of the cases and distance coefficients are negatively correlated, because the regressors are positively correlated ($r = 0.82$).

Delivery time data: standard errors

Using (34):

Coefficient	Estimate	Standard error
β_0 (intercept)	2.3412	1.0967
β_1 (cases)	1.6159	0.1707
β_2 (distance)	0.01438	0.003613

Coefficient of determination and adjusted R^2

Sums of squares

$$
\begin{aligned}
\mathrm{SS}_{\mathrm{T}} &= \sum_{i=1}^{n} (y_i - \bar{y})^2 = \mathbf{y}'\mathbf{y} - n\bar{y}^2, \\
\mathrm{SS}_{\mathrm{Res}} &= \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}, \\
\mathrm{SS}_{\mathrm{R}} &= \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - n\bar{y}^2, \\
\mathrm{SS}_{\mathrm{T}} &= \mathrm{SS}_{\mathrm{R}} + \mathrm{SS}_{\mathrm{Res}}.
\end{aligned} \tag{35}
$$

The decomposition follows from the orthogonality (19).

The coefficient of determination

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_{Res}}}{\mathrm{SS_T}}. \tag{36}$$

- The proportion of the variability in y explained by the regressors; $0 \leq R^2 \leq 1$.
- Delivery time data: $R^2 = 0.9596$.

Adding an irrelevant regressor

What happens to R^2 when a regressor is added to the model? Does it depend on whether the regressor is relevant?

Add z, 25 independent $N(0, 1)$ draws unrelated to the delivery time.

Model	p	R^2	MSE
cases	2	0.9305	17.484
cases, distance	3	0.9596	10.624
cases, distance, z	4	0.9597	11.109

R^2 increases slightly with the useless regressor, while the MSE goes up.

Adjusted R^2

$$R^2_{\text{adj}} = 1 - \frac{\text{SS}_{\text{Res}}/(n-p)}{\text{SS}_{\text{T}}/(n-1)} = 1 - \frac{\text{MS}_{\text{Res}}}{\text{MS}_{\text{T}}}, \tag{37}$$

$$= 1 - \frac{n-1}{n-p}(1 - R^2). \tag{38}$$

$\text{MS}_{\text{T}} = \text{SS}_{\text{T}}/(n-1)$ is the sample variance of y.

Why adjusted R^2 penalises useless regressors

- MS_{T} does not depend on the model, so by (37) R^2_{adj} increases if and only if $\mathrm{MS}_{\mathrm{Res}}$ decreases.
- Adding a regressor lowers SSE but also lowers the degrees of freedom $n - p$ in (33).
- The MSE decreases only if the drop in SSE outweighs the lost degree of freedom.

Adjusted R^2 in the example

Model	p	R^2	R^2_{adj}
cases	2	0.9305	0.9275
cases, distance	3	0.9596	0.9559
cases, distance, z	4	0.9597	0.9539

R^2_{adj} rises when the relevant regressor (distance) is added and falls when the irrelevant z is added.

Properties of adjusted R^2

From (38):

- For $p > 1$, $R^2_{\text{adj}} < R^2$ unless $R^2 = 1$.
- It can decrease when a regressor is added.
- It can be negative: $R^2_{\text{adj}} < 0$ whenever $R^2 < (p-1)/(n-1)$.

However,

- A high R^2 or R^2_{adj} does not mean the model is adequate: the assumptions must be checked with residual analysis.
- R^2_{adj} is a descriptive measure, not a hypothesis test.
- Formal comparison of models is done later with F-tests and information criteria.

SLR as a Special Case of MLR

SLR as a special Case of MLR

For SLR, $\mathbf{X} = [\mathbf{1} \;\; \mathbf{x}]$ and

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \qquad \mathbf{X}'\mathbf{y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}.$$

SLR as a special Case of MLR (cont.)

Since $n\sum x_i^2 - (\sum x_i)^2 = nS_{xx}$,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}.$$

The second entry of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is

$$\hat{\beta}_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{nS_{xx}} = \frac{nS_{xy}}{nS_{xx}} = \frac{S_{xy}}{S_{xx}},$$

which is (1).

SLR variance from the MLR covariance matrix

In equation (23) we found that $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

Using the inverse computed for SLR, with $\sum x_i^2 = S_{xx} + n\bar{x}^2$:

$$\mathrm{Var}(\hat{\beta}_1) = \sigma^2 \frac{n}{nS_{xx}} = \frac{\sigma^2}{S_{xx}},$$

$$\mathrm{Var}(\hat{\beta}_0) = \sigma^2 \frac{\sum x_i^2}{nS_{xx}} = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right),$$

$$\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2 \frac{\sum x_i}{nS_{xx}} = -\sigma^2 \frac{\bar{x}}{S_{xx}}.$$

These are the familiar SLR results.

Summary

Summary

- Model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $\mathrm{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ (4).
- OLS: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ (10); $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is a projection onto $\mathrm{col}(\mathbf{X})$.
- $\hat{\boldsymbol{\beta}}$ is unbiased with $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ (23), and is BLUE.
- $\hat{\sigma}^2 = \mathrm{SS}_{\mathrm{Res}}/(n-p)$ is unbiased for σ^2 (33).
- R^2 never decreases when regressors are added; R^2_{adj} (37) penalises the loss of degrees of freedom.

Next lecture

- Inference on the regression coefficients: t-tests and confidence intervals.
- Test for significance of regression (F-test).
- Reading: Montgomery, Peck & Vining, Chapter 3.

Proof of the Gauss–Markov Theorem

Gauss–Markov: setup of the proof

Any linear estimator can be written

$$\tilde{\boldsymbol{\beta}} = [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}]\mathbf{y}$$

for some $p \times n$ matrix $\mathbf{D}$. Its expectation is $\boldsymbol{\beta} + \mathbf{D}\mathbf{X}\boldsymbol{\beta}$, so unbiasedness for every $\boldsymbol{\beta}$ requires

$$\mathbf{D}\mathbf{X} = \mathbf{0}. \tag{39}$$

Gauss–Markov: the variance comparison

$$
\begin{aligned}
\operatorname{Var}(\tilde{\boldsymbol{\beta}}) &= \sigma^2\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}\big]\big[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'\big] \\
&= \sigma^2\big[(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{D}\mathbf{X})' \\
&\qquad + \mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}\mathbf{D}'\big] \\
&= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{D}\mathbf{D}',
\end{aligned}
$$

where the last step uses (39).

Gauss–Markov: conclusion

- $\mathbf{a}'\mathbf{D}\mathbf{D}'\mathbf{a} = \|\mathbf{D}'\mathbf{a}\|^2 \geq 0$ for every $\mathbf{a}$, so $\mathbf{D}\mathbf{D}'$ is positive semidefinite.
- Hence $\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}})$ is positive semidefinite, and $\mathrm{Var}(\mathbf{c}'\tilde{\boldsymbol{\beta}}) \geq \mathrm{Var}(\mathbf{c}'\hat{\boldsymbol{\beta}})$ for every $\mathbf{c}$.
- Equality holds only for $\mathbf{D} = \mathbf{0}$, i.e. $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$.

Questions?